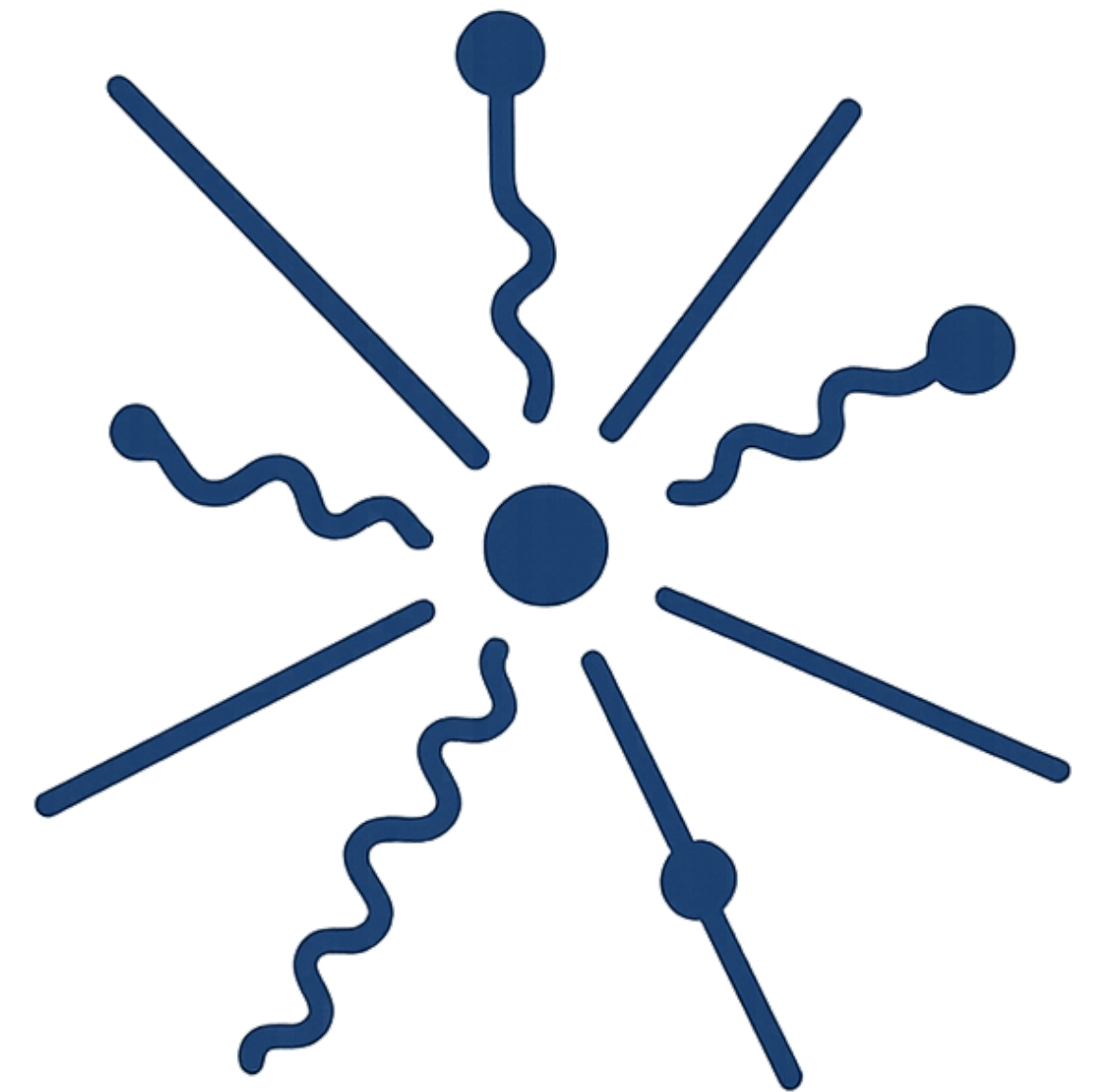
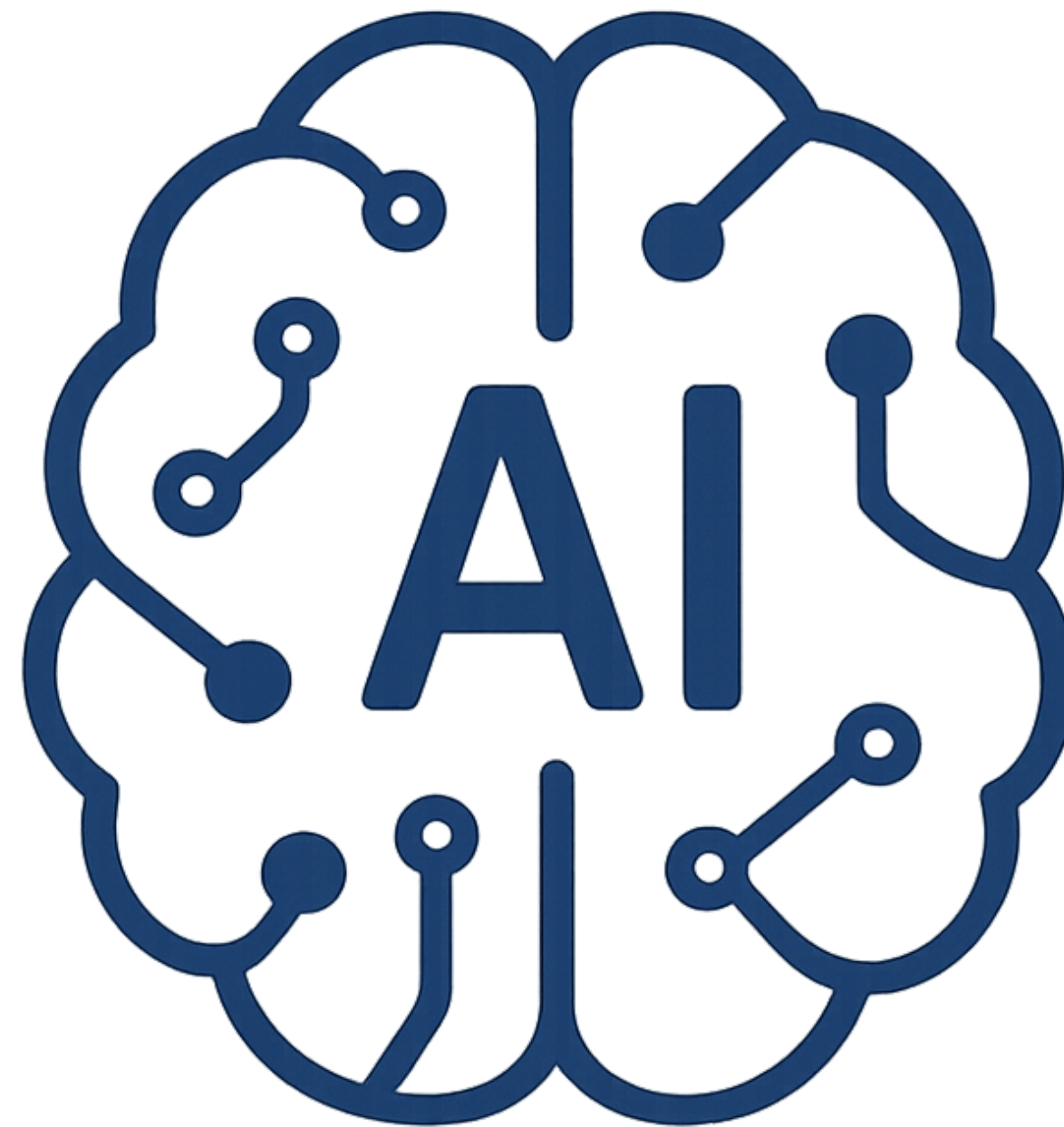
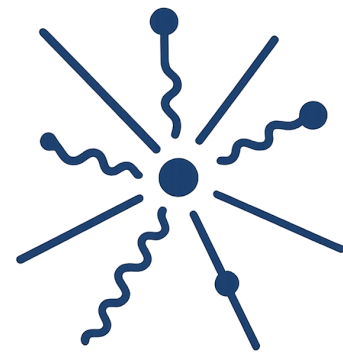
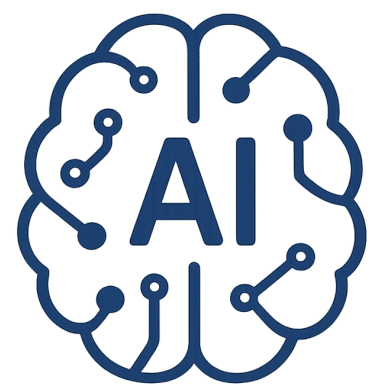


Introduction to

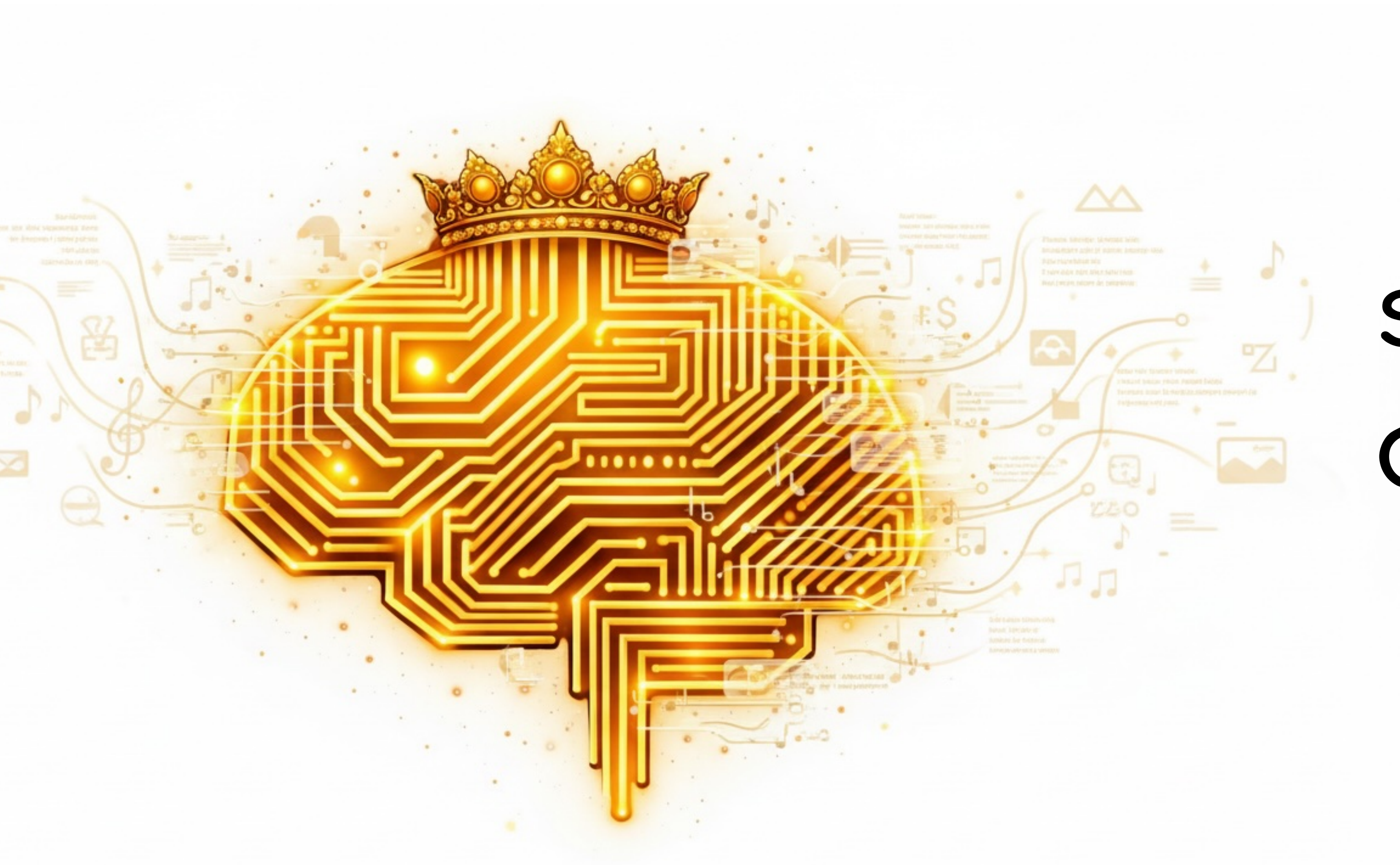
AI-Driven HEP

S. A. Fard - School of Physics (IPM)



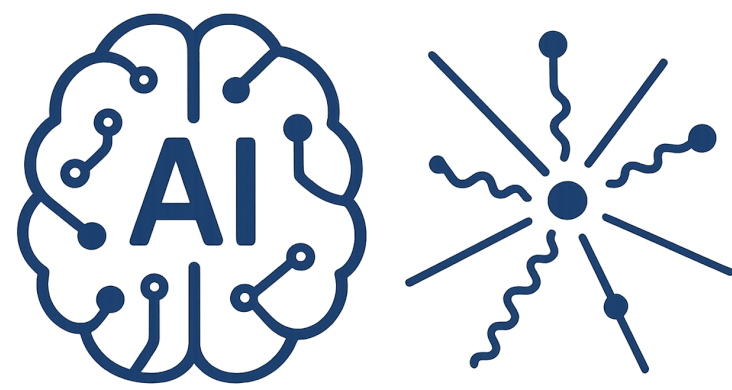


AI-Driven HEP 22



Session 22

Generative AI



AI-Driven HEP 22

Reference

- [1] Bishop, Christopher M., and Hugh Bishop. Deep learning: Foundations and concepts. Springer Nature, 2023.
- [2] Simon J.D. Prince, Understanding Deep Learning, The MIT Press (2025), <https://udlbook.github.io/udlbook/>
- [3] Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks, <https://arxiv.org/abs/1703.10593>
- [4] Normalizing Flows: An Introduction and Review of Current Methods, <https://arxiv.org/abs/1908.09257>
- [5] Normalizing Flows for Probabilistic Modeling and Inference, <https://arxiv.org/pdf/1912.02762v2>,
- [6] Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks, <https://arxiv.org/abs/1511.06434v2>

<https://generated.photos/>

<https://physics.ipm.ac.ir/~ansarifard/>





Do you know these people ?



Do you know these people ?



They are not real

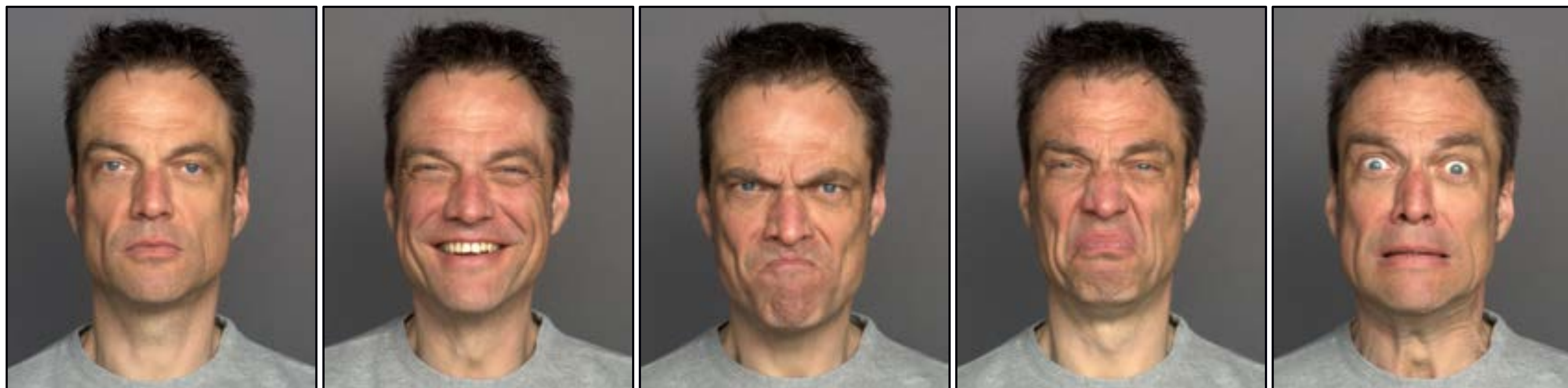
Do you know these people ?

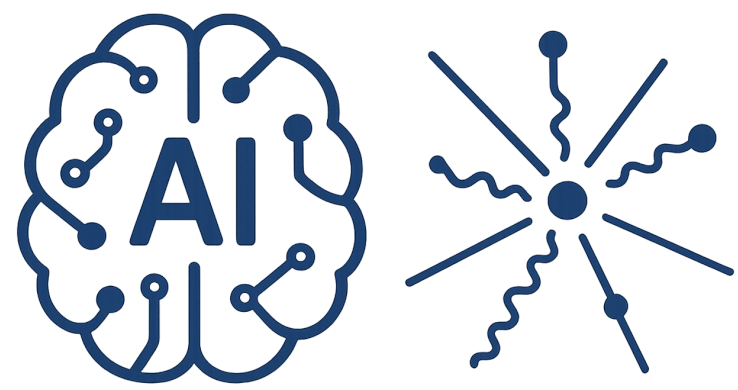


They are not real

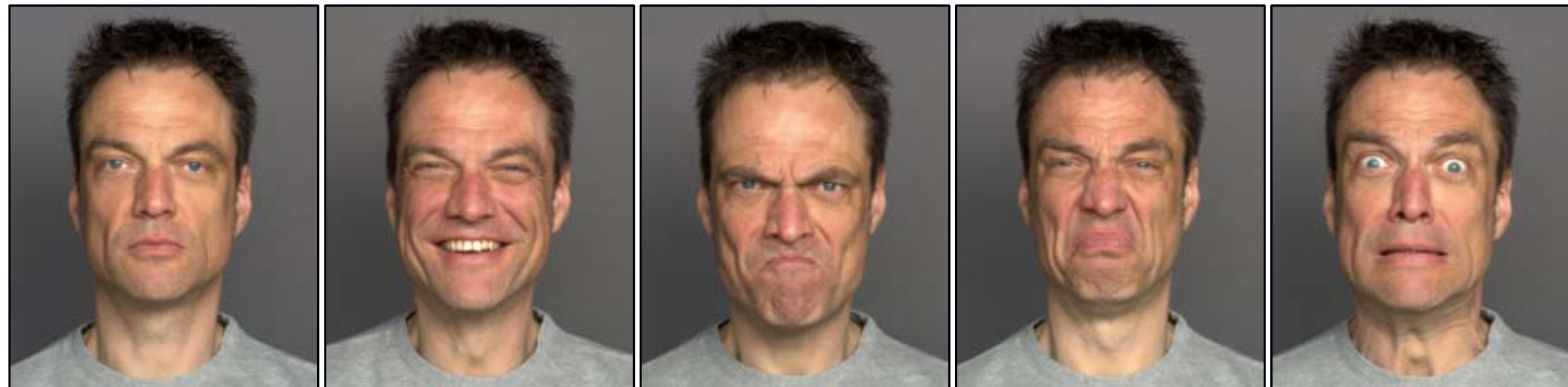
What is real ?

Latent Space

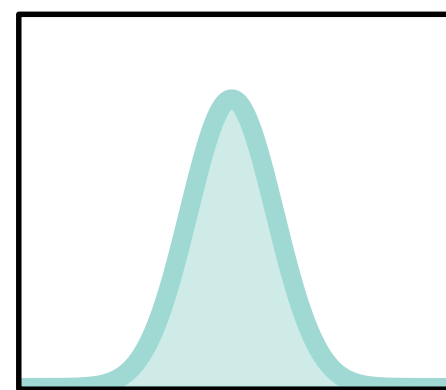




Latent Space



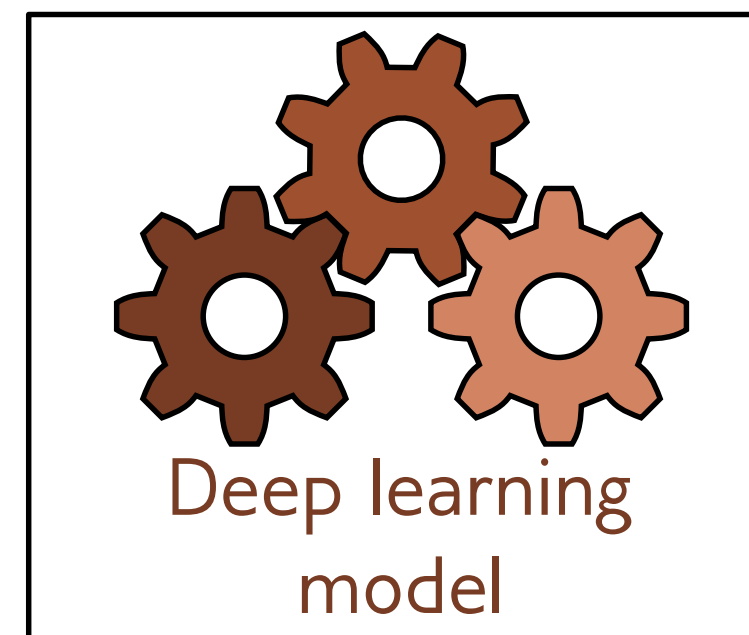
Normal
distribution



Latent
variables

$$\begin{bmatrix} -0.5 \\ 0.1 \\ 1.2 \\ -0.6 \\ \vdots \end{bmatrix}$$

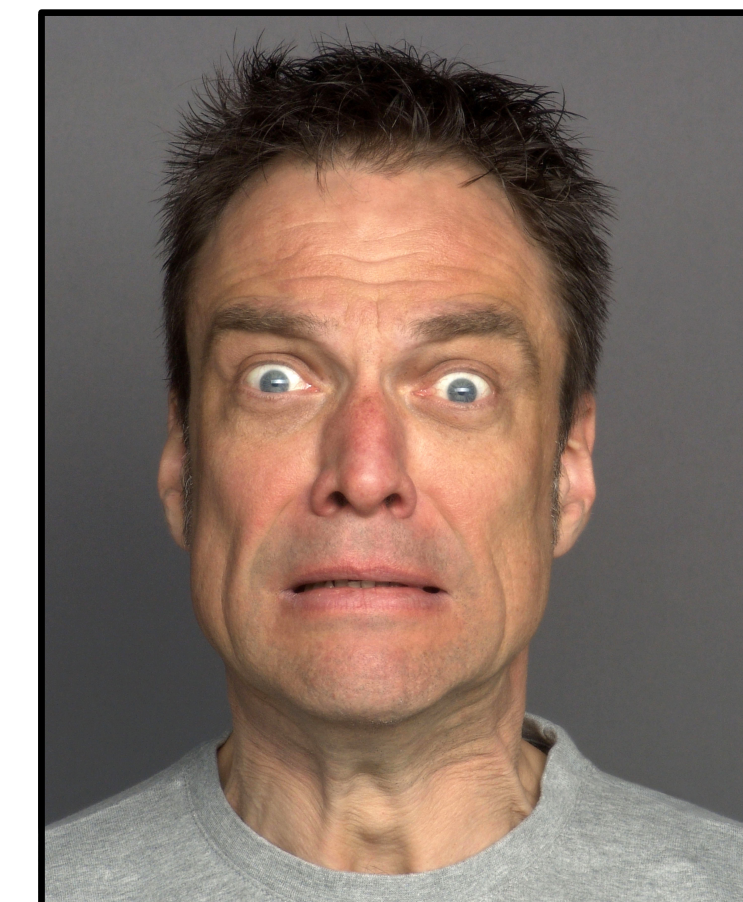
Model



Model output

$\begin{bmatrix} 110 \\ 109 \\ 110 \\ 108 \\ 109 \\ 110 \\ 110 \\ 110 \\ 109 \\ \vdots \end{bmatrix}$

Real world output





Unsupervised Learning

Generate sample to

manipulate

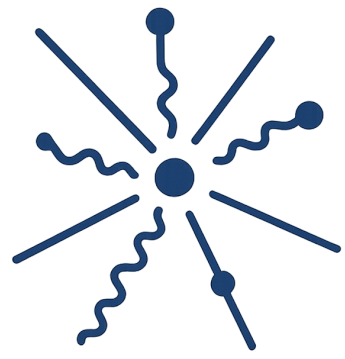
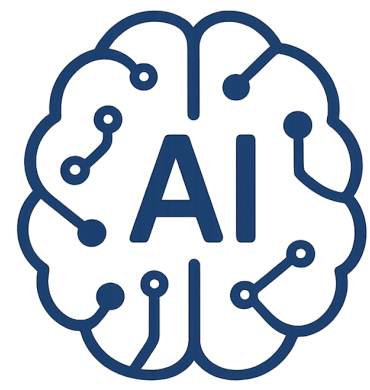
denoise

interpolate

compress

internal structure

distinguish outliers



Data Space

5

(N, D)

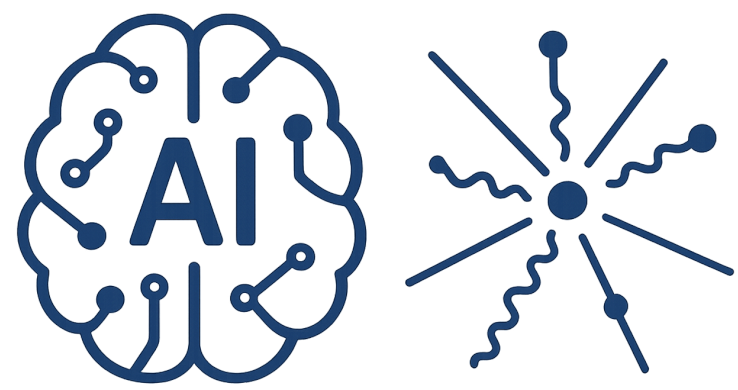
dimensions of natural representation

- i.i.d
- Sequential

Image Patches

Video Frames

Event Particles



Data Space



(N, D)

dimensions of natural representation

- i.i.d
- Sequential

Image Patches

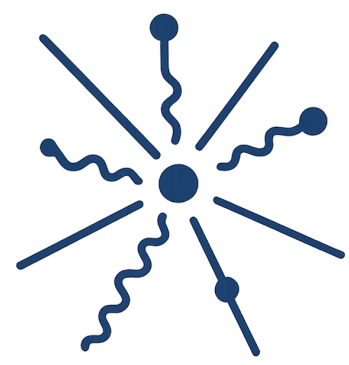
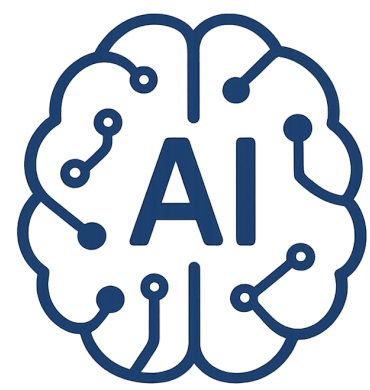
Video Frames

Event Particles

Multimodal

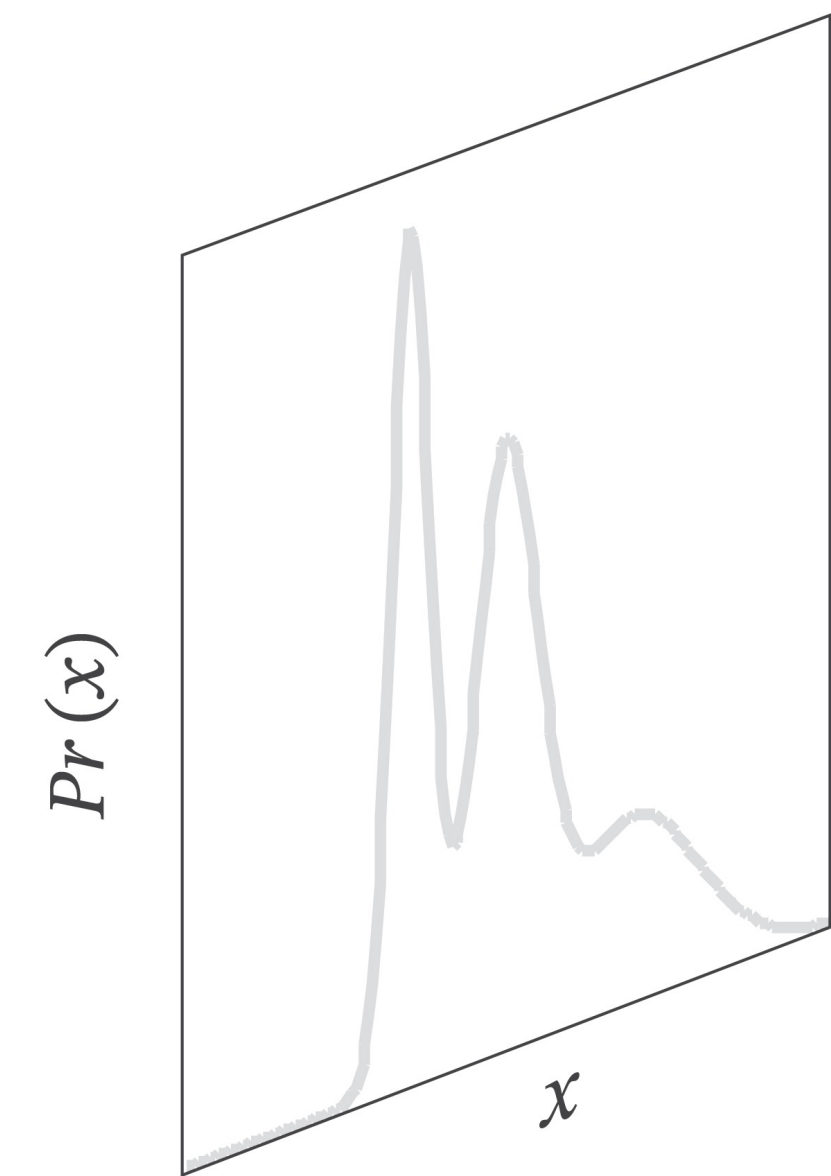
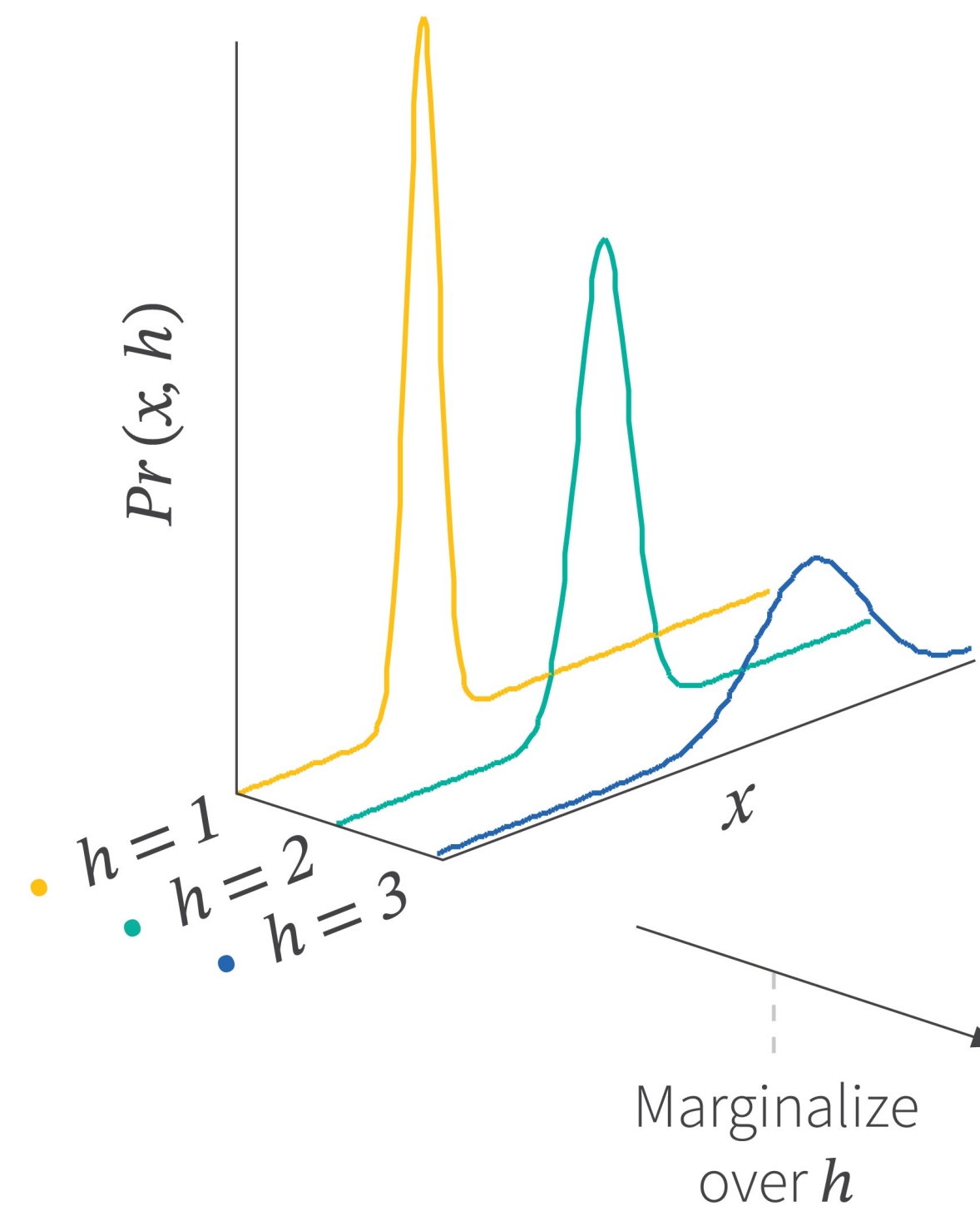
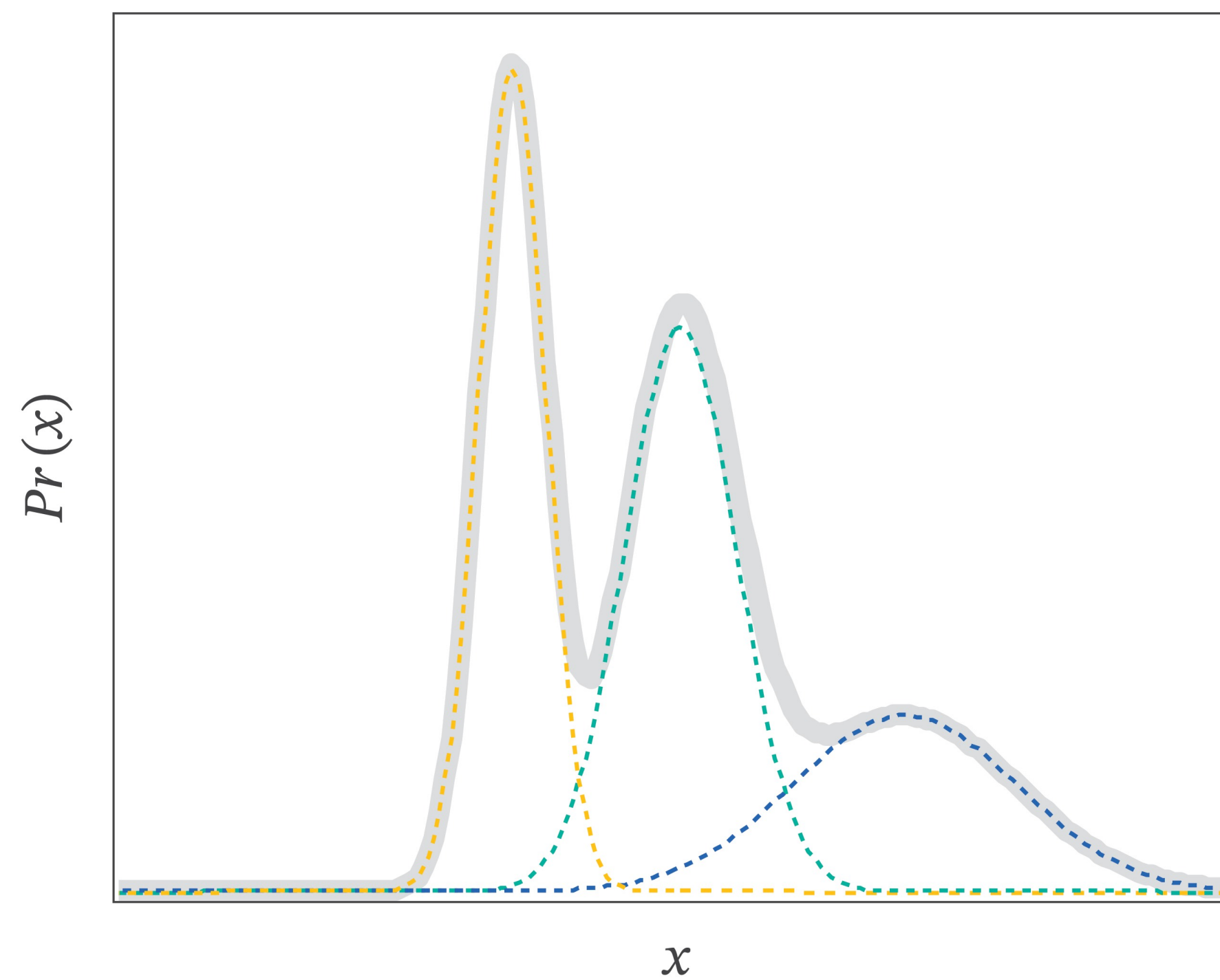
Multimodal The probability distribution over the data space has multiple modes (peaks)

$$p(x) = \sum_{k=1}^K \pi_k p_k(x)$$



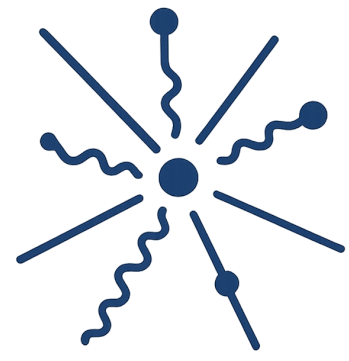
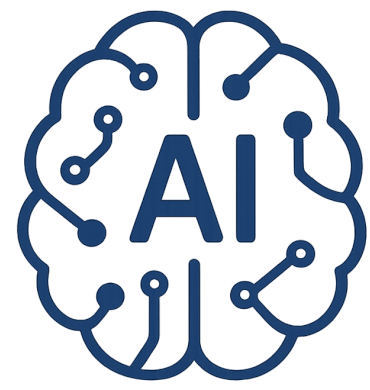
Mixture of Gaussians

continuous observed data and discrete latent



Linear Latent Model



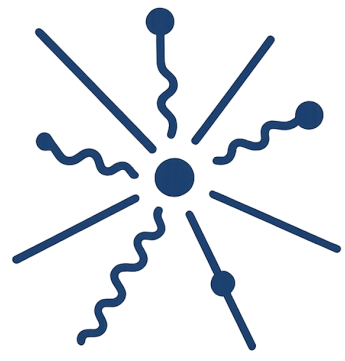
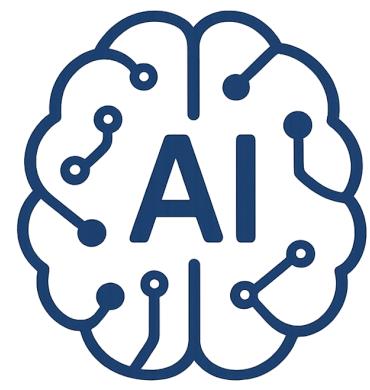


Linear Latent Model



$$\mu = \frac{1}{N} \sum_n X_n$$

$$X_n \in \mathbb{R}^D, \quad D = 28 \times 28$$



Linear Latent Model



$$\tilde{X}_n = X_n - \mu,$$

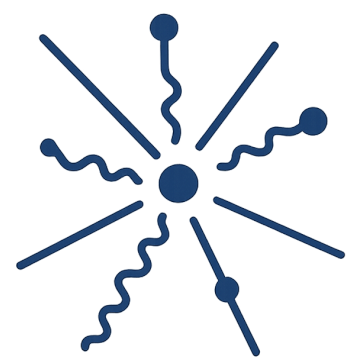
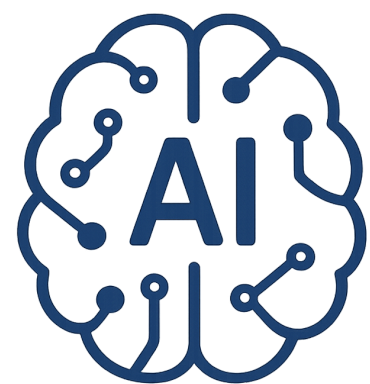
$$\Sigma = \frac{1}{N} \sum_n \tilde{X}_n^T \tilde{X}_n$$

$$\Sigma \in \mathbb{R}^{D \times D}, \quad D = 784$$

$$\mu = \frac{1}{N} \sum_n X_n$$

$$X_n \in \mathbb{R}^D, \quad D = 28 \times 28$$





Linear Latent Model



$$\mu = \frac{1}{N} \sum_n X_n$$

$$X_n \in \mathbb{R}^D, \quad D = 28 \times 28$$

$$\tilde{X}_n = X_n - \mu,$$

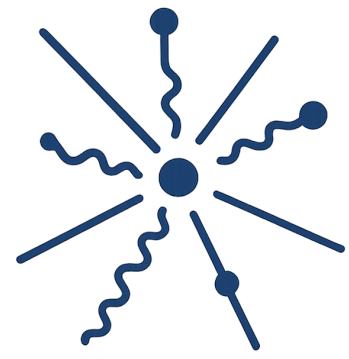
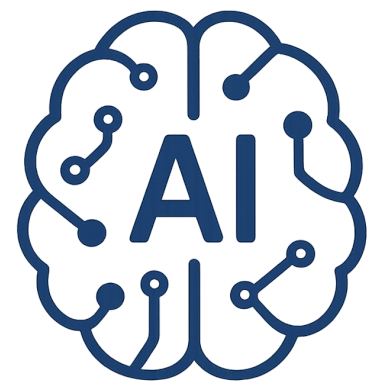
$$\Sigma = \frac{1}{N} \sum_n \tilde{X}_n^T \tilde{X}_n$$

$$\Sigma \in \mathbb{R}^{D \times D}, \quad D = 784$$



$$\Sigma v = \lambda v$$

$$\lambda_1 > \lambda_2 > \dots > \lambda_D$$

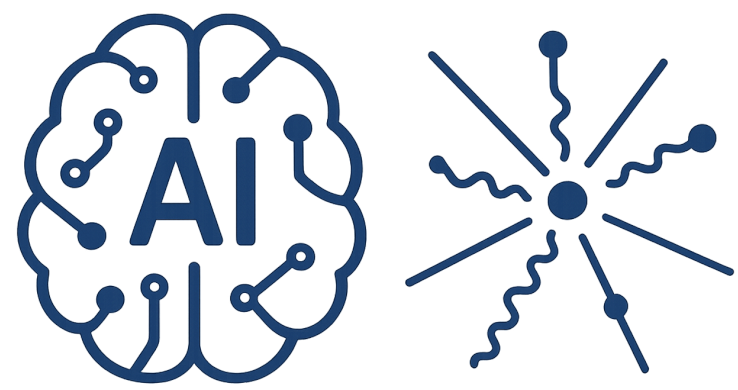


Principal Component Analysis

5

Generate images from latent variable

$$x(z_1) = \mu + z_1 v_1 \quad \lambda_1 \sim 8$$



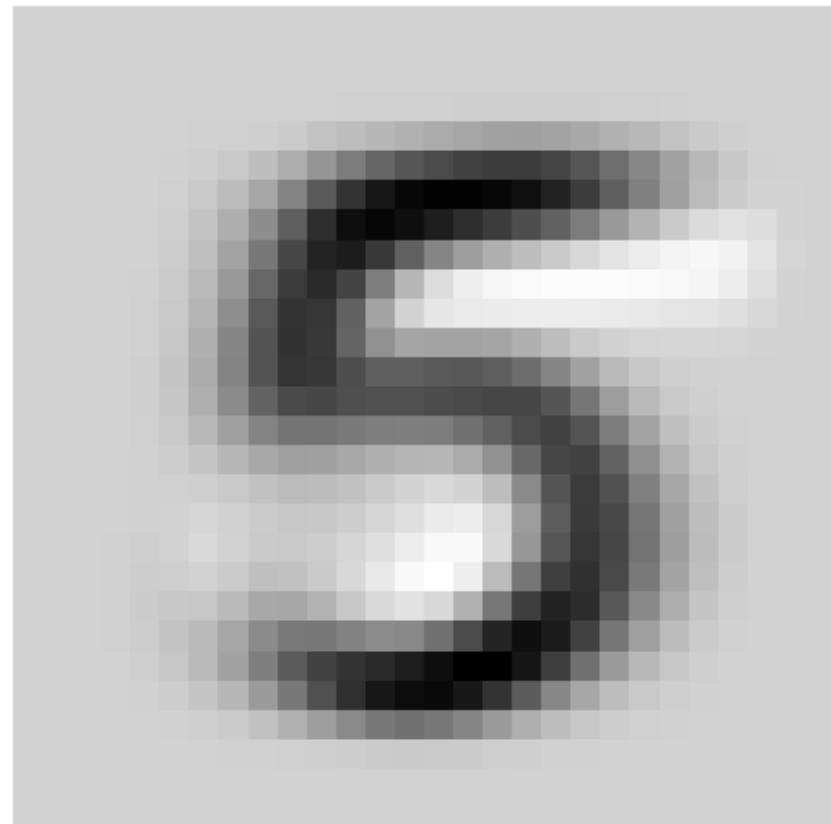
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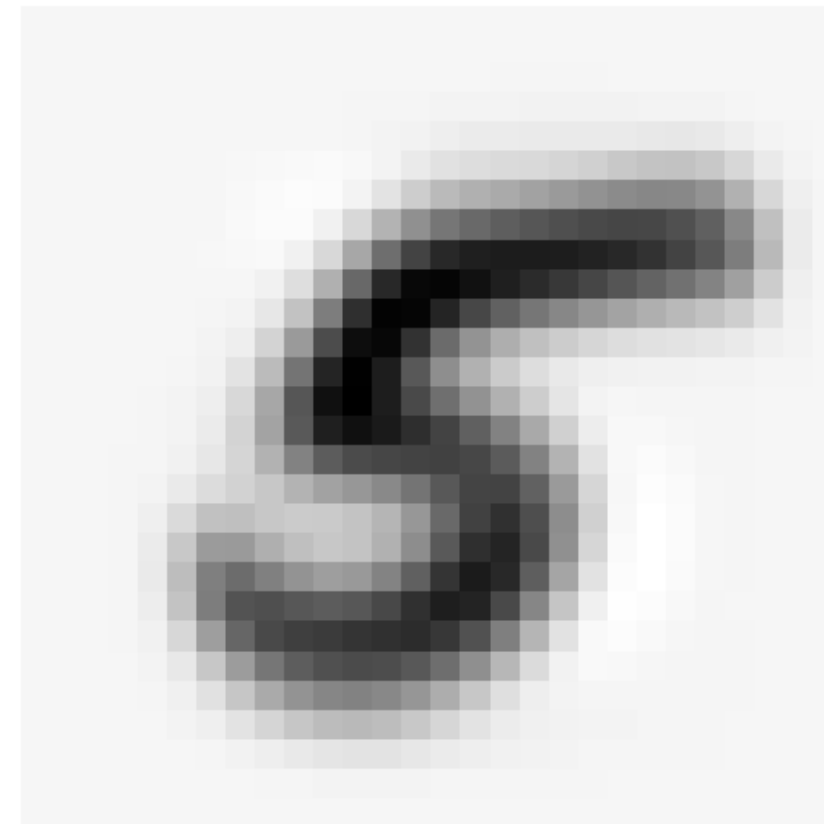
$z_1 = -5.64$



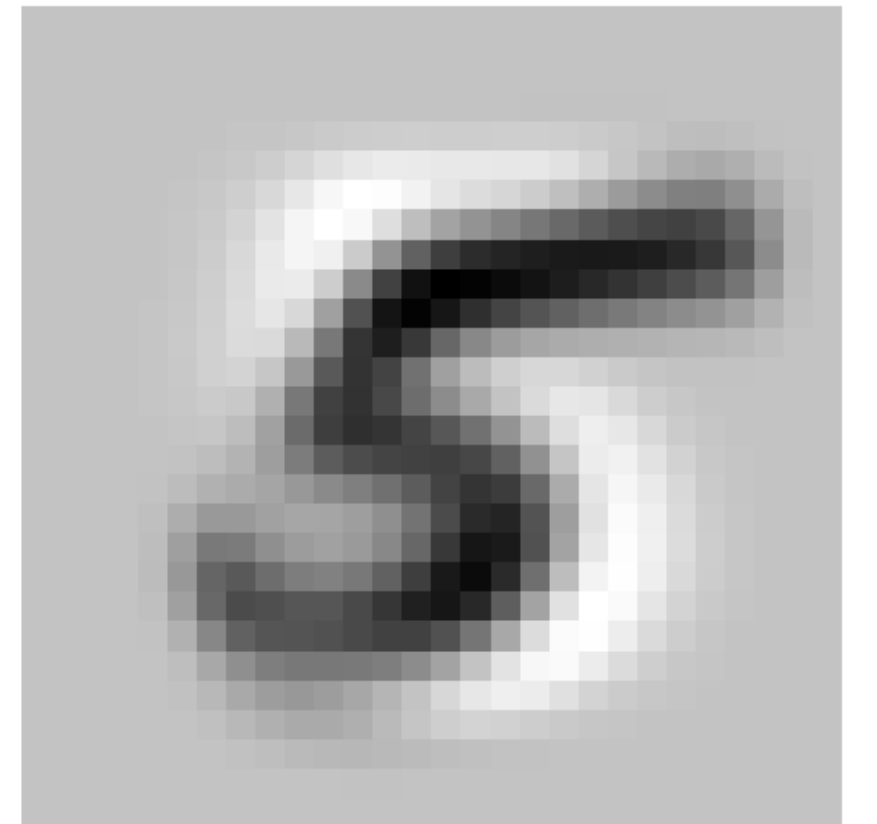
$z_1 = -2.82$



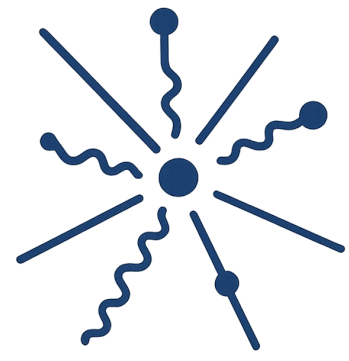
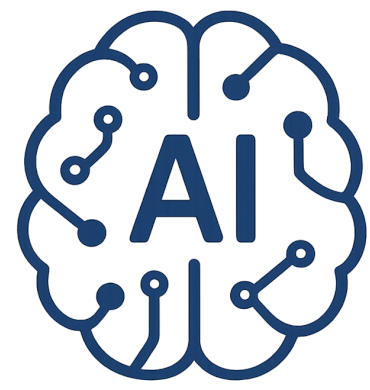
$z_1 = 2.82$



$z_1 = 5.64$



$$z_1 \sim \sqrt{\lambda_1}$$



AI-Driven HEP 22

PCA



In principle Each image x can be written as

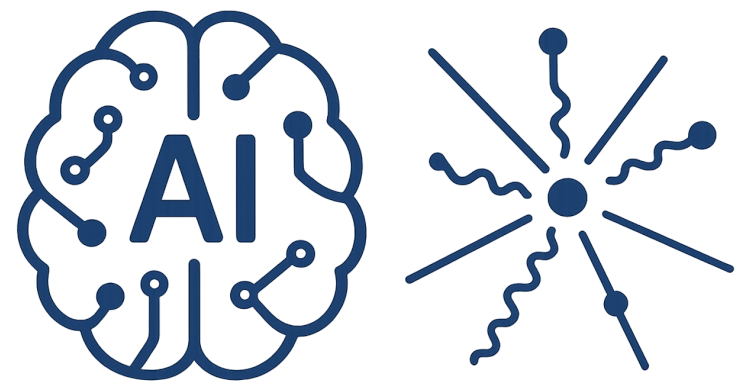
$$x = \mu + \sum_{i=1}^D z_i v_i$$

Latent vector

$$z = (z_1, \dots, z_D) \in \mathbb{R}^D$$

$$\Lambda = \text{diag}(\lambda_1, \dots, \lambda_D)$$

$$V = [v_1, \dots, v_D]$$



PCA



In principle Each image x can be written as

$$x = \mu + \sum_{i=1}^D z_i v_i$$

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$$z = (z_1, \dots, z_D) \in \mathbb{R}^D$$

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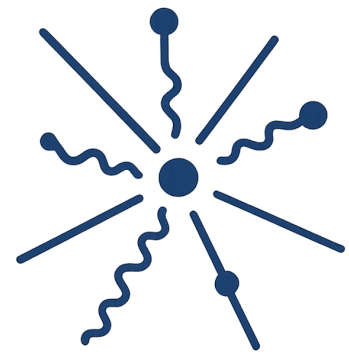
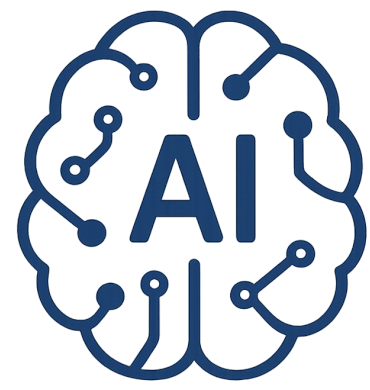
$$V = [v_1, \dots, v_D]$$

Eigenvalues typically decay

$$\lambda_1 \gg \lambda_2 \gg \dots \gg \lambda_D \approx 0$$

K latent dimensions is ~ 50

$$x = \mu + \sum_{i=1}^K z_i v_i$$



Data Whitening

5

Features are uncorrelated

Each feature has unit variance

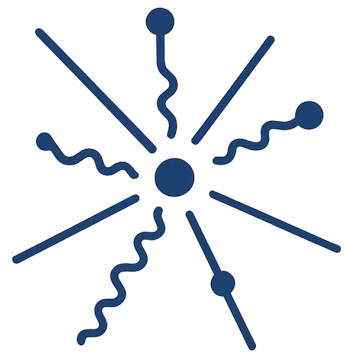
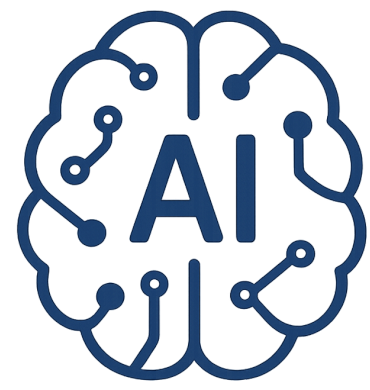
$$\mathbb{E}[x] = 0, \quad \text{Cov}(x) = I$$

$$x_{\text{white}} = \Lambda^{-1/2} V^{\top} (x - \mu)$$

Data Whitening

5 $x_{\text{white}} = \Lambda^{-1/2} V^T (x - \mu)$

$$\mu = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \Sigma = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}$$



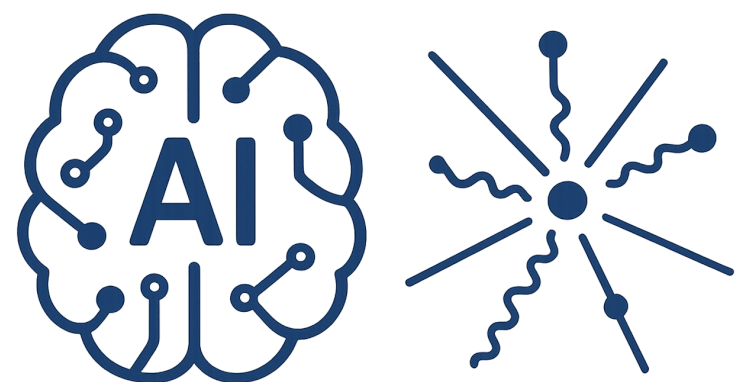
Data Whitening

5

$$x_{\text{white}} = \Lambda^{-1/2} V^T (x - \mu)$$

$$\mu = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \Sigma = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}$$

$$\Lambda = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix} \quad V = \begin{pmatrix} \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{pmatrix}$$



Data Whitening

5

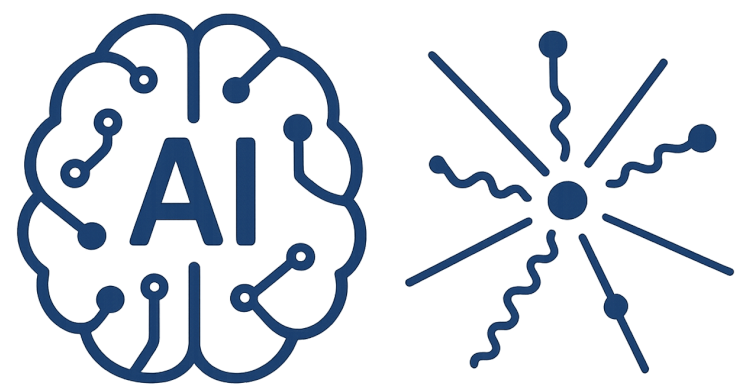
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$$x = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$$V^T (x - \mu) = \begin{pmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} \frac{6}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{pmatrix}$$



Data Whitening

5

$$x_{\text{white}} = \Lambda^{-1/2} V^T (x - \mu)$$

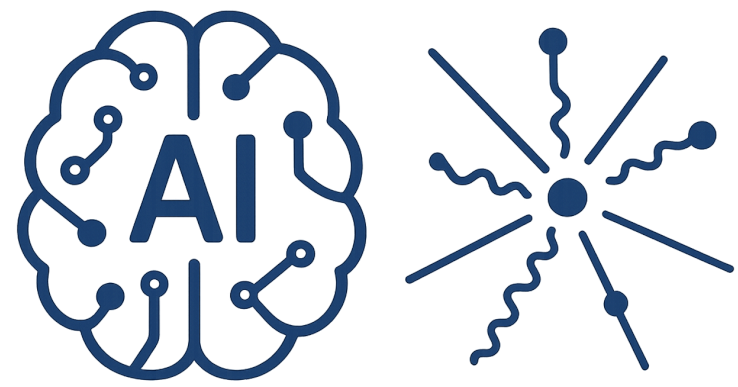
$$x = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

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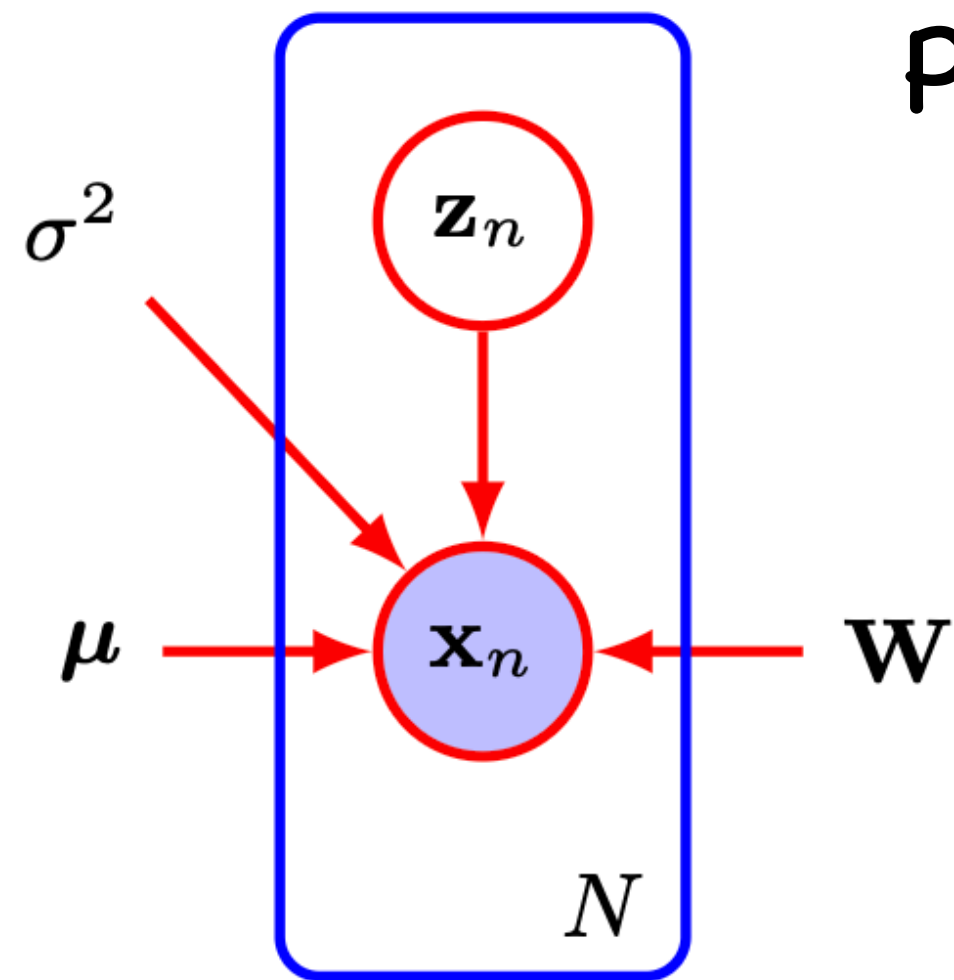
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$$\Lambda = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix} \quad V = \begin{pmatrix} \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{pmatrix}$$

$$x_{\text{white}} = \begin{pmatrix} \frac{1}{\sqrt{5}} & 0 \\ 0 & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{6}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{pmatrix} = \begin{pmatrix} \frac{6}{5} \\ \frac{2}{\sqrt{10}} \end{pmatrix}$$



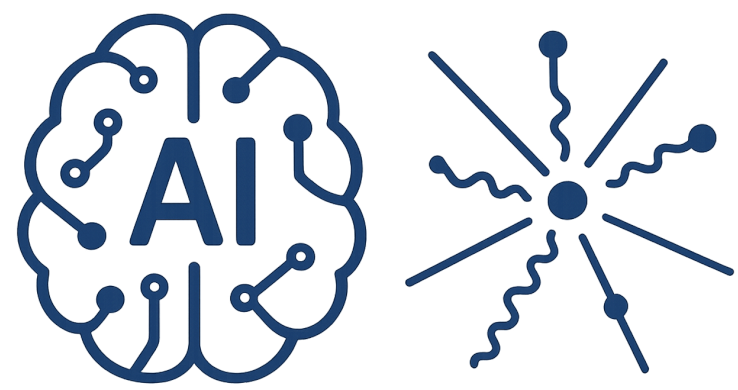
Probabilistic PCA



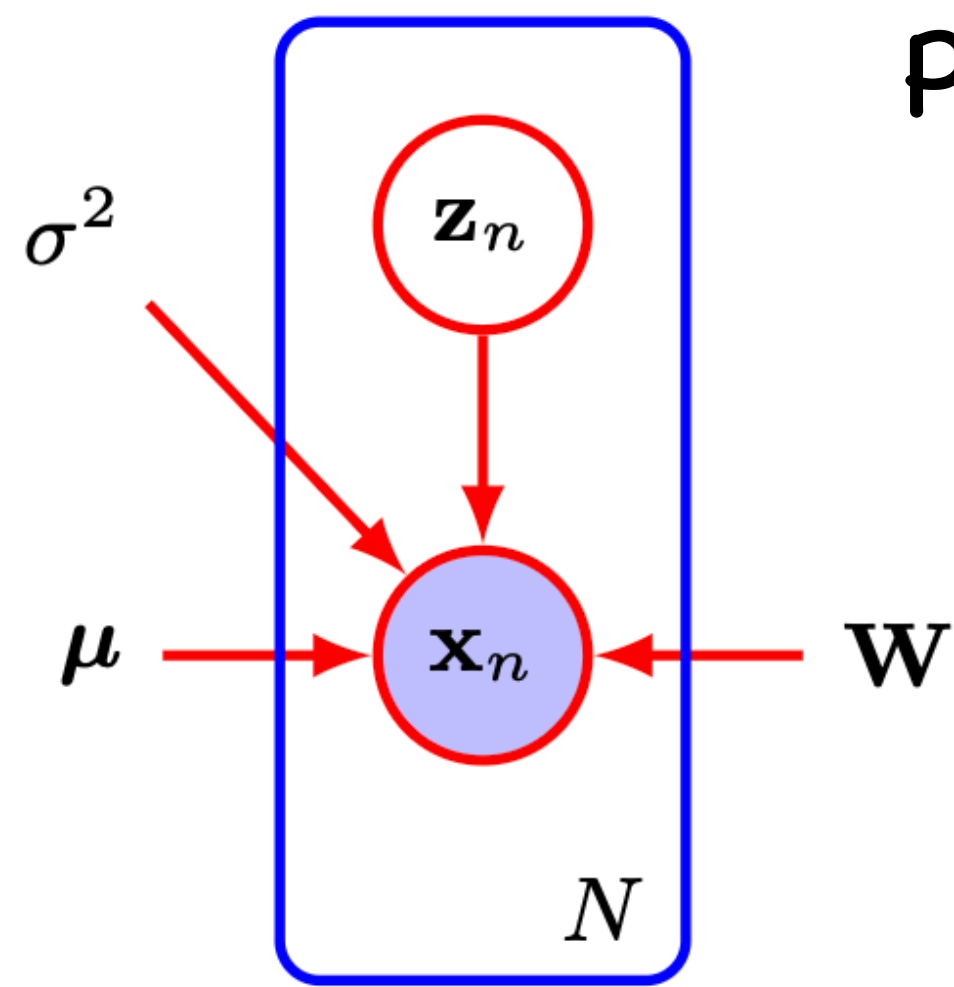
Prior distribution over the latent variable

$$p(\mathbf{z}) \sim \mathcal{N}(\mathbf{0}, I)$$

$$\mathbf{z} \in \mathbb{R}^K, \quad I \in \mathbb{R}^{K \times K}$$



Probabilistic PCA

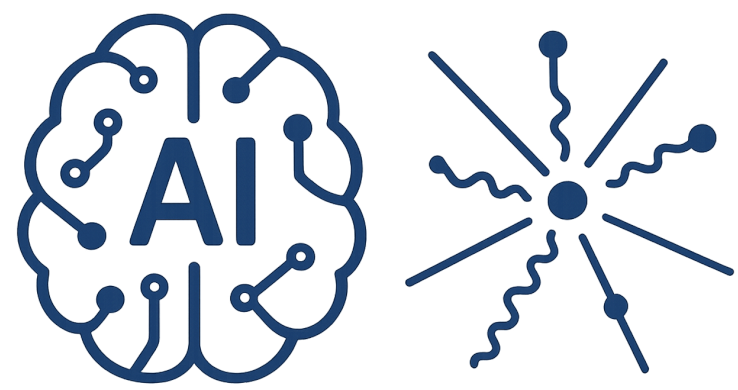


Prior distribution over the latent variable

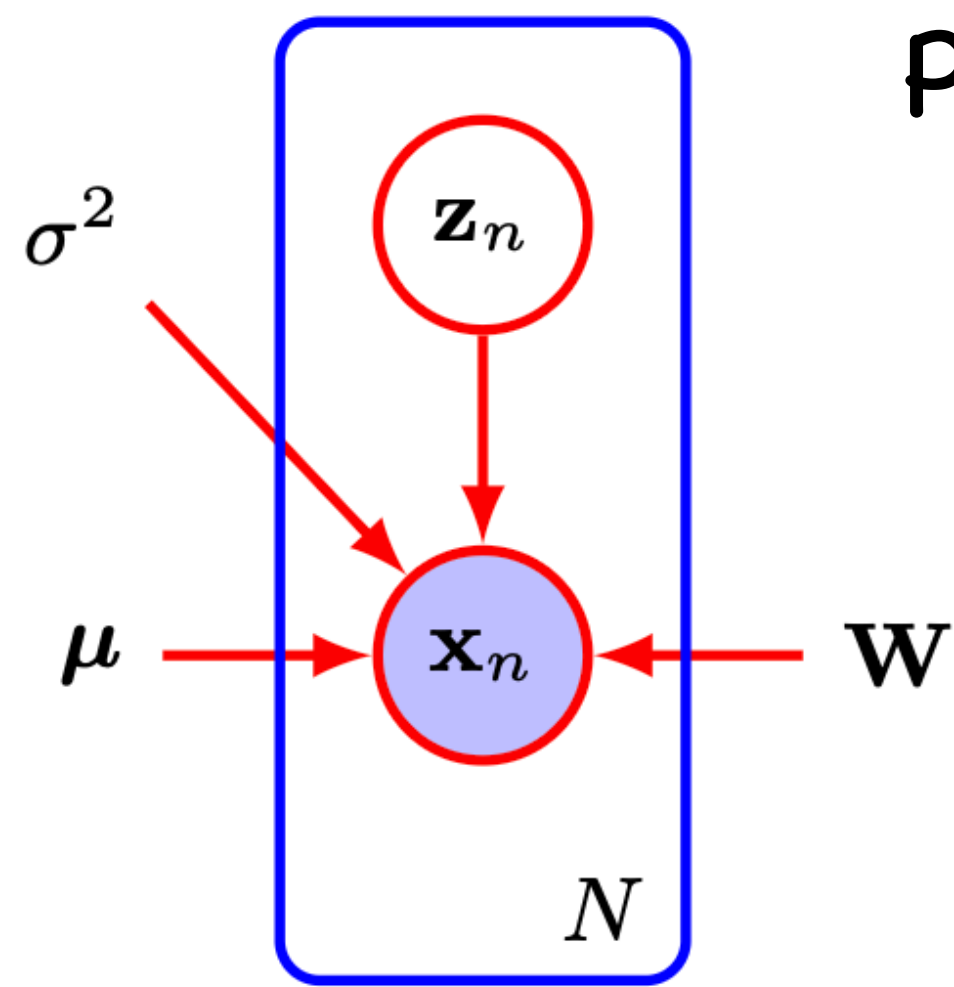
$$p(z) \sim \mathcal{N}(0, I) \quad z \in \mathbb{R}^K, \quad I \in \mathbb{R}^{K \times K}$$

Conditional distribution (linear decoder)

$$p(x|z) \sim \mathcal{N}(x | Wz + \mu, \sigma^2 I) \quad W \in \mathbb{R}^{D \times K}, \quad I \in \mathbb{R}^{D \times D}$$



Probabilistic PCA



Prior distribution over the latent variable

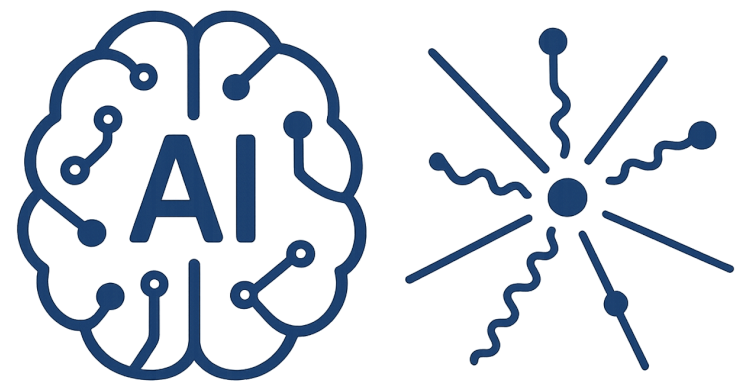
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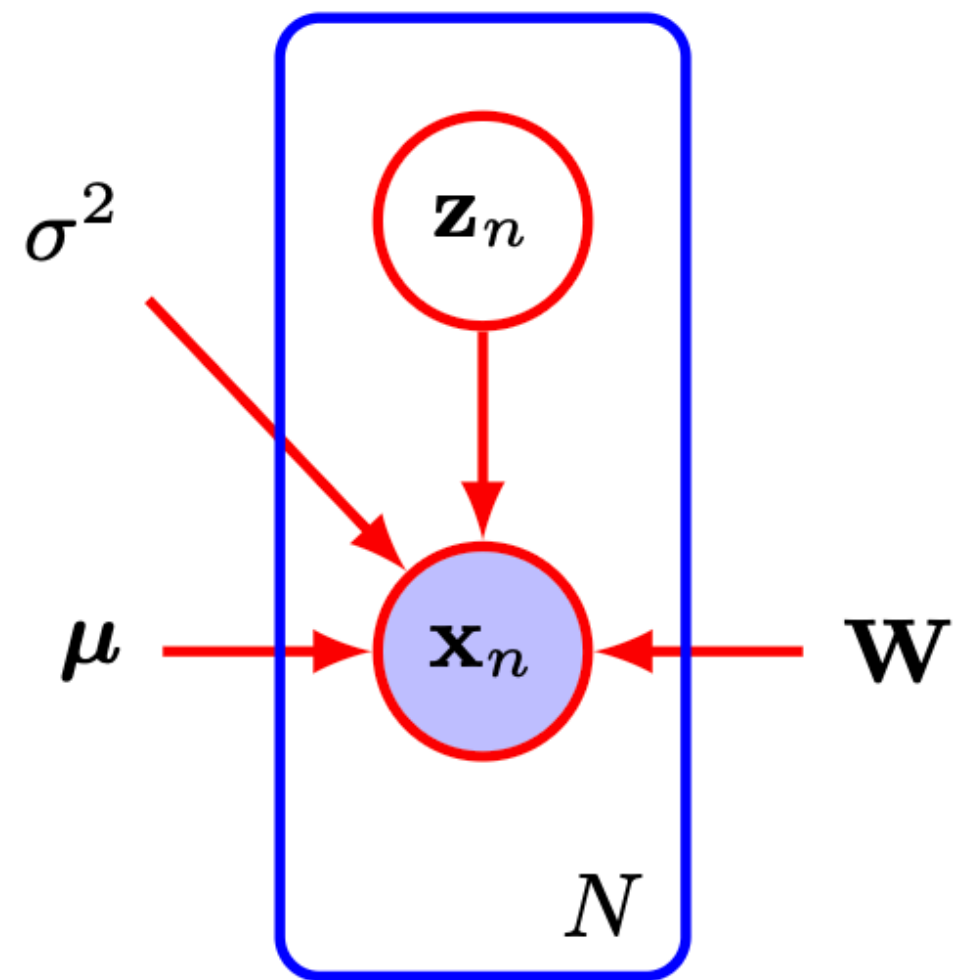
$$p(x) = \int p(x|z)p(z) dz \quad \text{Marginal (predictive) distribution}$$

$$p(x) = \mathcal{N}(x | \mu, WW^\top + \sigma^2 I)$$



Probabilistic PCA

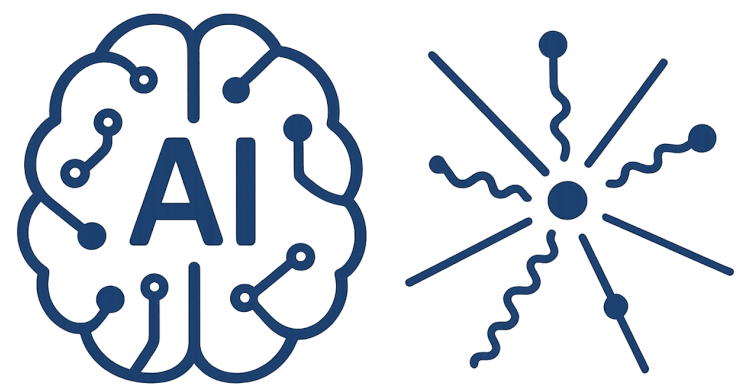
Posterior Distribution



$$p(z | x) = \mathcal{N}(z | M^{-1}W^{\top}(x - \mu), \sigma^2 M^{-1})$$

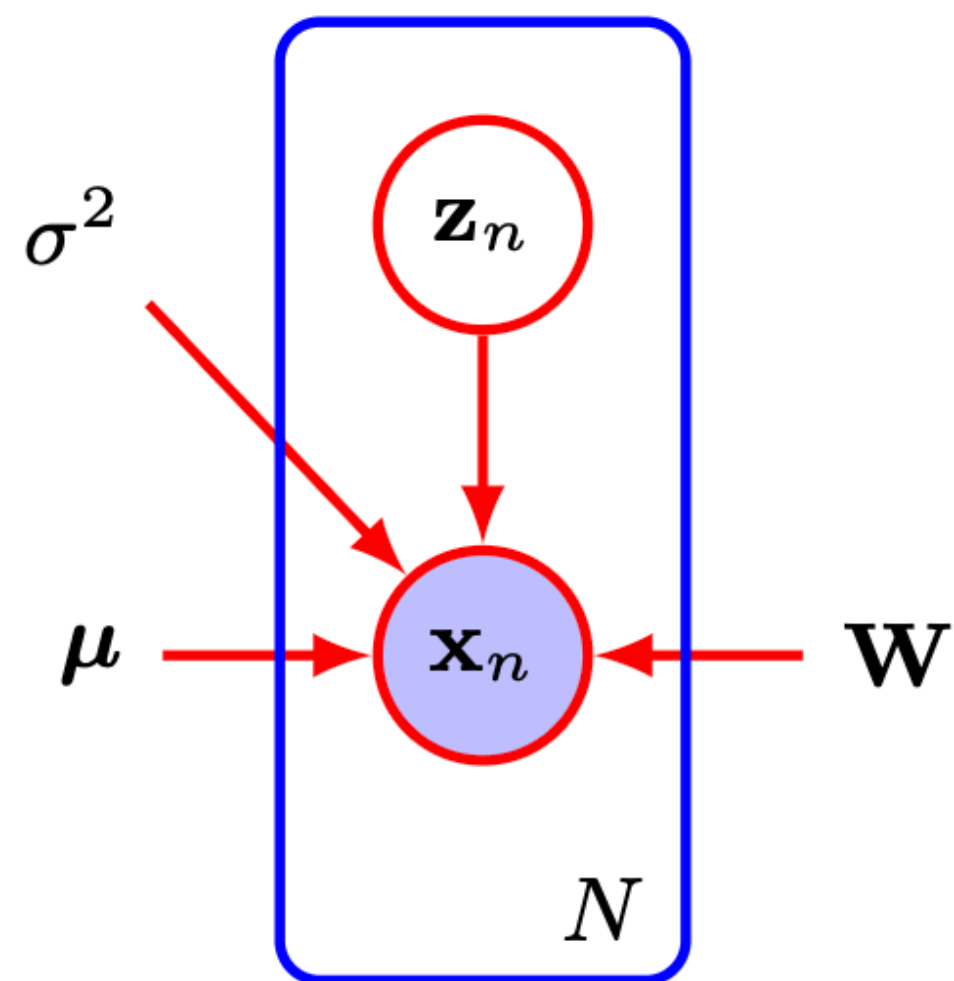
$$M = W^{\top}W + \sigma^2 I$$

Latent vector z
given observation x



Probabilistic PCA

Posterior Distribution



$$p(z | x) = \mathcal{N}(z | M^{-1}W^{\top}(x - \mu), \sigma^2 M^{-1})$$

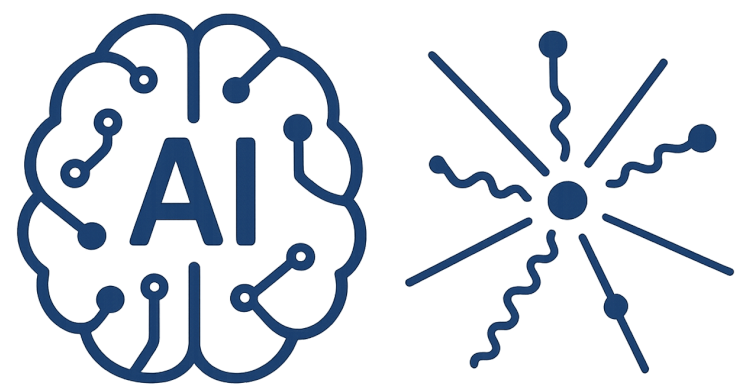
$$M = W^{\top}W + \sigma^2 I$$

Latent vector z
given observation x

Reduce to deterministic latent PCA in $\sigma^2 \rightarrow 0$

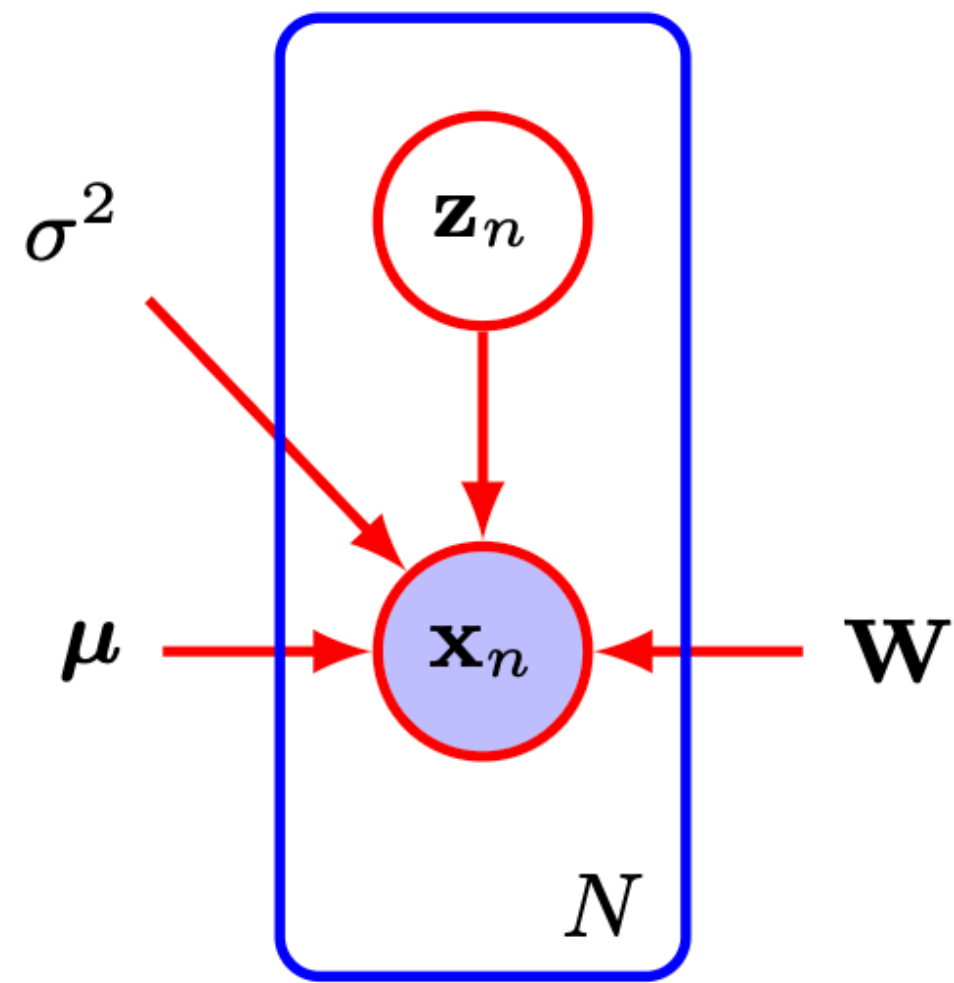
$$z \approx (W^{\top}W)^{-1}W^{\top}(x - \mu)$$

Whitened PCA



Probabilistic PCA

Posterior Distribution



$$p(z | x) = \mathcal{N}(z | M^{-1}W^{\top}(x - \mu), \sigma^2 M^{-1})$$

Latent vector z
given observation x

$$M = W^{\top}W + \sigma^2 I$$

Noise Estimation

average variance associated with the
discarded dimensions

$$\sigma_{ML}^2 = \frac{1}{D - K} \sum_i \lambda_i$$

Reduce to deterministic latent PCA in $\sigma^2 \rightarrow 0$

$$z \approx (W^{\top}W)^{-1}W^{\top}(x - \mu)$$

Whitened PCA

Data splits to PCA latent space and noise space